Fundamental of STL Part 2

Saturday, March 14, 2026 9:57 PM

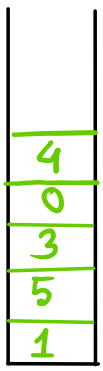
MAP

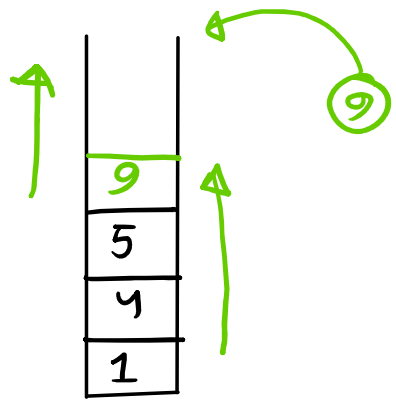
```
map<int, int> KeyValupMap;

KVM[1] = 2;
KVM[5] = 3;
KVM[6] = 10;
```

insert( )
erase( )
count( )
} O(logN)

Stack → LIFO → Last In First Out

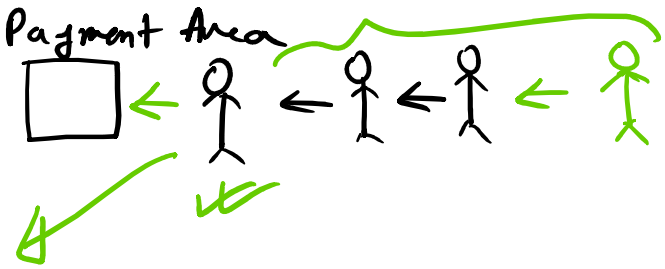
1, 5, 3, 4, 5, 0, 4, 7



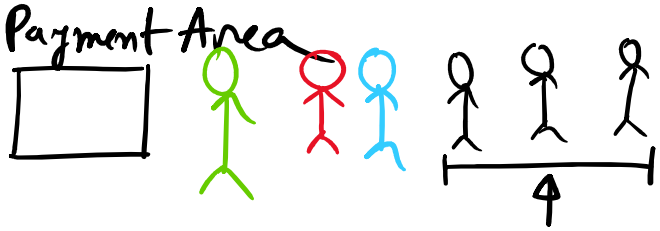


monotonic Stack

Queue - FIFO → First in First out



priority Queue



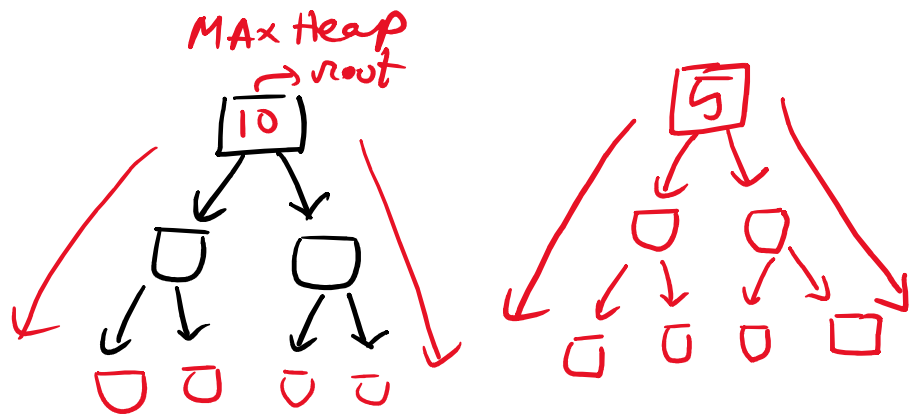
10 7 7 6 5

int

10 7 7 6 5

Heap Tree / Heap Sort

push() } → log(n)
pop() }

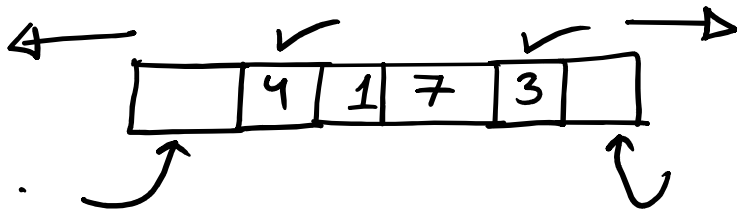


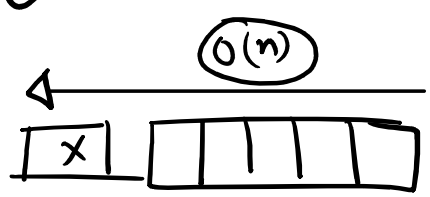
Deque

Dequeue → pop

Enqueue → push

→ Double-ended Queue





sort(——) → Into - sort

TC: O(N log N)

SC: O(log N)

① Quick Sort ← >100

② Insertion Sort ← <100

③ ——— ———

Lower Bound and Upper Bound

① Sorted ✓

0	1	2	3	4	5	6
1,	2,	3,	3,	3,	5	100

-1

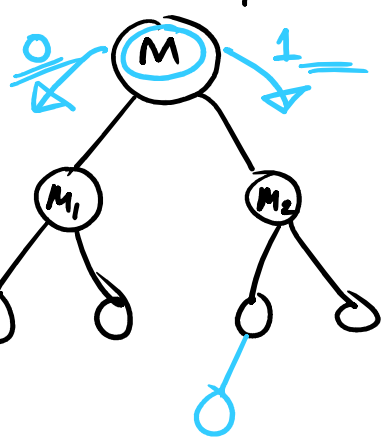
lb(4) = 5
ub(4) = 5

lowerbound(3) = 2

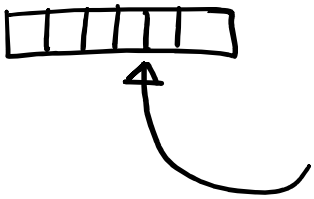
upperbound(3) = 5

ub(100) = 6
lb(-1) = 0

insert()



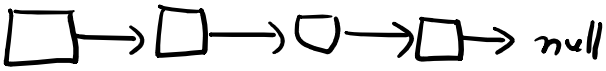
nodeValue	insert value
0,	1



Stack memory ✗

Heap memory ✓

Dynamic Memory Allocate, Heap memory



2x